Quantum information, communication, and computation

Problem Set 1

Semester 2, 2026

Due **5 pm** on **Friday** 21 Aug.

Submit your problem set (typed, or hand written and photographed/scanned) via Blackboard.

Answers should include detailed comments and reasoning. A grade up to 5 can be obtained by answering only questions in Part A. Higher grades require all questions to be answered (with clear, well reasoned arguments showing originality and insight). Refer to the Grading Criteria document for details.

Problem 1.1
Part A

We looked at how Pauli matrices help describe rotations of qubits on the Bloch sphere. We will now prove this rotation property by using the Bloch-vector representation of qubits. Note the notation $\mathbf{r}\bullet\boldsymbol{\sigma} \equiv r_x\sigma_x + r_y\sigma_z + r_z\sigma_z$.

(a) Prove for an arbitrary qubit state $|\psi\rangle = (\cos(\frac{\theta}{2}), e^{i\phi}\sin(\frac{\theta}{2}))$, the density operator defined as $\rho = |\psi\rangle\langle\psi|$ simplifies into $\rho = \frac{1}{2}(\mathbf{I}_2 + \mathbf{r}\bullet\boldsymbol{\sigma})$. Note that $\mathbf{r} = (r_x, r_y, r_z)$ is termed the Bloch vector and is a convenient representation of qubits on the Bloch sphere.

(b) Prove the Euler identity for Pauli matrices: $e^{i\theta\cdot\mathbf{r}\bullet\boldsymbol{\sigma}} = \mathbf{I}_2\cos(\theta) + i(\mathbf{r}\bullet\boldsymbol{\sigma})\sin(\theta)$. *Hint: use the Taylor expansion/definition for the matrix exponential and notice the values of σ_x^2, σ_y^2 and σ_z^2.*

(c) Show that Pauli matrices satisfy:

$$\sigma_x\sigma_y = -\sigma_y\sigma_x = i\sigma_z$$
$$\sigma_y\sigma_z = -\sigma_z\sigma_y = i\sigma_x$$
$$\sigma_z\sigma_x = -\sigma_x\sigma_z = i\sigma_y$$

or equivalently: $\sigma_p\sigma_q = \delta_{pq}\mathbf{I}_2 + i\sum_{r=1}^{3}\varepsilon_{pqr}\sigma_r$.

(d) Prove for $m \in x, y, z$, the operator to rotate a qubit state on the Bloch sphere by angle α around the axis m is $R_m(\alpha) = e^{-i\frac{\alpha}{2}\sigma_m}$. *Hint: ρ is an operator, think about how $R_m(\alpha)$ transforms ρ into ρ' via a sandwich. Then calculate ρ' due to a rotation about axis m and simplify as much as possible using trigonometric identities and the previous results. Finally, insert $m = x, y, z$ and separately show that the resulting Bloch vector (r_x, r_y, r_z) is transformed via rotations (c.f. the form of 3D axial rotation matrices).*

Part B - Advanced

(a) In the previous section, the rotations were proved for the axes. Now prove that any general rotation can be given as discussed in the lectures: $R_{\mathbf{n}}(\varphi) = e^{i\frac{\varphi}{2}\cdot(\mathbf{n}\bullet\boldsymbol{\sigma})}$. *Hint: use the Euler identity and show that it leads to a general rotation transformation; that is, pick a general rotation representation such as $ZYZYZ$.*

(b) Prove the no-cloning theorem (that is, given an arbitrary quantum state $|\psi\rangle$ and an known initial state $|\psi_0\rangle$, there is no unitary operator $\hat{Q}$ where $\hat{Q}(|\psi\rangle\otimes|\psi_0\rangle) = |\psi\rangle\otimes|\psi\rangle$). Comment on the particular axioms of quantum mechanics that enforce the no-cloning theorem.

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Problem 1.2

In a circuit-based quantum computer (e.g. spin qubits, atomic qubits, superconducting qubits, ion-trapped qubits etc.), gates are implemented by carefully manipulating the Hamiltonian parameters over time. The following exercise schematically represents the basic considerations/intuition used when operating such devices.

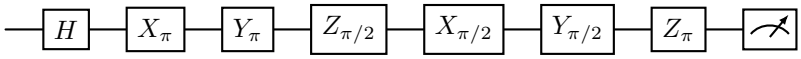
Assume that each qubit in this quantum computer follows the Hamiltonian $\mathbf{H} = \hbar(t_c\sigma_x + \Delta\sigma_z)$ and the values of t_c, and Δ can be independently and precisely controlled/set over time (note that the units of these parameters are in angular frequency). We note that the system constrains $0 \leqslant t_c/(2\pi) \leqslant 5\,\text{GHz}$ and $0 \leqslant \Delta/(2\pi) \leqslant 40\,\text{GHz}$.

(a) Describe how we can achieve qubit rotations about the x or z axes by setting appropriate values for the Hamiltonian parameters and waiting. Calculate the fastest possible X_π and Z_π gates.

(b) Describe how we may implement Y_π. *Hint: can a y-rotation be described as a concatenation of x-axis and z-axis rotations?*

(c) Describe how the Hadamard gate H can be implemented.

(d) The machinery setting the pulses has a time-resolution of 1 ns. That is, it can only set values for t_c and Δ in 1 ns time-steps. While respecting all given constraints, implement a timing-diagram for the values of t_c and Δ to implement the following circuit:



Note that the qubit must be in a Hamiltonian state that will not change the measurement result during the measurement phase in the final step.

Part B - Advanced

Continuing from the previous section, let's introduce 2-qubit gates. Consider 2 qubits. We have the ability to tune the interaction via a binary pulse on parameter C, where we set C to 0 or 1 to completely turn the two-qubit interaction off or on. When the interaction is off, the two qubits each obey the Hamiltonian outlined in the previous section. To keep things simple, when the interaction is on, the effective Hamiltonian leaves the first qubit's Hamiltonian the same, while only perturbing the second qubit via:

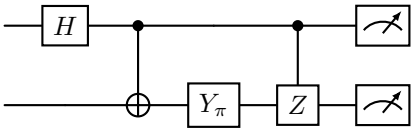
$$\mathbf{H}_{\text{Qubit 2}} = \begin{cases} \hbar(t_c\sigma_x + (\Delta + \delta_0)\sigma_z) & \text{Qubit 1 is } |0\rangle \\ \hbar(t_c\sigma_x + (\Delta + \delta_1)\sigma_z) & \text{Qubit 1 is } |1\rangle \end{cases}$$

where $\delta_0/(2\pi) = 1\,\text{MHz}$ and $\delta_1/(2\pi) = 51\,\text{MHz}$.

(a) Describe how a CZ gate can be performed by setting t_c, Δ and C. What is the fastest CZ gate that can be performed? Note that a compensating phase gate may be required (on either/both qubits) to ensure that no relative phase accumulation occurs between the qubits as that would imply a residual z-rotation gate (remember, global phases across qubits do not affect their dynamics).

(b) Describe how we may implement a CX gate.

(c) The machinery setting the analogue and digital pulses still has a time-resolution of 1 ns. While respecting all given constraints, implement a timing-diagram for the values of t_c, Δ and C to implement the following circuit:



Problem 1.3

Part A
For this problem, we will consider the Schmidt decomposition and the entangled state $|\Psi\rangle$ given by,

$$|\Psi\rangle = \frac{1}{\sqrt{2}} \left(|\alpha_1\rangle_A |0\rangle_B + |\alpha_2\rangle_A |1\rangle_B \right)$$

where

$$|\alpha_1\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle$$
$$|\alpha_2\rangle = \cos\theta|0\rangle - \sin\theta|1\rangle.$$

(a) What is the physical significance of the Schmidt decomposition?

(b) Is the Schmidt decomposition unique? Justify your answer.

(c) What is the Schmidt decomposition of the state $|\Psi\rangle$

(d) What is the reduced density matrix for subsystem B?

Part B
Suppose that qubit A of the entangled state $|\Psi\rangle$ is with Alice, and qubit B is with Bob. If Alice can determine that her qubit is in the state $|\alpha_1\rangle$ and not in the state $|\alpha_2\rangle$, she can conclude that Bob's qubit is in the state $|0\rangle$. Similarly, if she can determine that her qubit is in the state $|\alpha_2\rangle$ and not in the state $|\alpha_1\rangle$, she can conclude that Bob's qubit is in the state $|1\rangle$. Failure of such unambiguous state discrimination will result to Bob's qubit being in some other state (i.e. not $|0\rangle$ or $|1\rangle$).

What is the upper bound on the probability for unambiguous state discrimination? *(Hint: Use the no-signalling theorem.)*

Problem 1.4

Part A
Suppose that Alice wishes to teleport an unknown pure state of a qubit to Bob of the form:

$$|a\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle.$$

Unfortunately, their entangled state generator can produce only non-maximally entangled states given by $|AB\rangle = \cos\beta|0\rangle_A|1\rangle_B - \sin\beta|1\rangle_A|0\rangle_B$.

(a) For each of the Alice's Bell basis measurement, list the states obtained by Bob.

(b) Find an expression for the fidelity of the qubit teleported as function of θ and ϕ.

(c) What is the average fidelity of teleportation for this unknown qubit?

Part B
Suppose that Alice wishes to teleport an unknown pure state of a qubit to Bob. Unfortunately, their entangled state generator is not reliable, so they can only share an imperfect entangled state in the form of a Werner state $\hat{\rho}_W = \frac{p}{4}\hat{\mathrm{I}}\otimes\hat{\mathrm{I}} + (1-p)\,|\Psi^-\rangle\langle\Psi^-|$. Calculate the fidelity of the qubit that Bob receives, with respect to the qubit that Alice wished to send.